

Natural Language Processing Simplified

Natural language processing (NLP) is the technology that enables computers to recognise human language, precisely. It is the way to program computers to process and analyse large amounts of natural language data which could be text, content or speech. NLP is a component of artificial intelligence.

Artificial Intelligence

Enables computers to mimic human behaviour

Machine Learning

Technique for realizing AI

Deep Learning

Subject of ML, which uses algorithms inspired by function of human brain

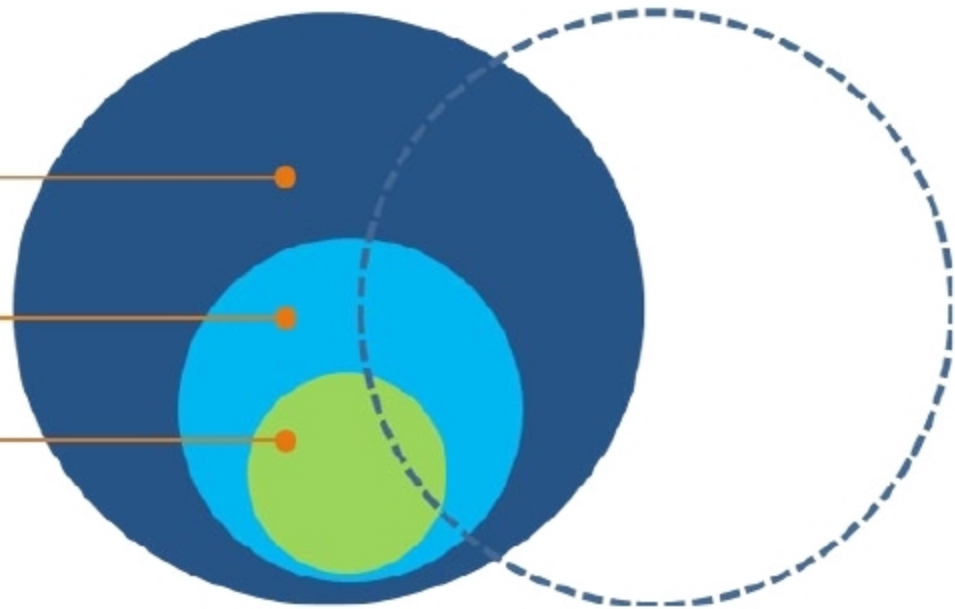


Figure 1: How NLP is related to AI, ML and DL

Hierarchically, NLP is considered as a subset of machine learning (ML), while NLP and ML both fall under the larger category of artificial intelligence (AI).

The primary objective of NLP is to combine artificial intelligence (AI) and computational linguistics so that computers and humans can talk to each other, seamlessly. To communicate with humans, a program must apprehend syntax (grammar), semantics (phrase meaning), morphology (tense) and pragmatics (conversation). NLP enables computer programs to comprehend unstructured content by utilising AI and ML to make derivations and give context to language, similar to how human brains function.

Common everyday applications of NLP

The application of NLP was less impressive earlier. But today, it has advanced considerably to yield good

outcomes for some common NLP tasks, such as named entity recognition (NER), parts of speech (PoS) tagging and sentiment analysis – all of which leverage neural network models to outperform traditional approaches. Today, NLP is everywhere even if we don't realise it, covering everything from search and online translation to spam filters and spell checking. In this post, we look at the four areas where natural language processing is becoming a reality.

Machine translation: In the last few decades, the number of non-English speaking Internet users has drastically increased all over the world, which means a big chunk of the data feed to the Internet is non-English. To stay competitive, companies in international markets, especially in new and growing markets, have jumped to translating their texts into many

more languages. Business houses have started introducing new applications and improved machine learning NLP models to automate translations.

Automatic summarisation:

Information overload is a real problem when we need to access a specific, important piece of information from a huge knowledge base. Automatic summarisation is especially relevant when used to provide an overview of a news item or a set of blog posts, while avoiding redundancy from multiple sources and maximising the diversity of content obtained.

Sentiment analysis: Sentiment analysis is extremely useful in social media monitoring, as it allows us to gain an overview of the wider public opinion on certain topics. Retailers use NLP applications such as sentiment analysis to identify opinions and sentiments online, and factor these

